

Tracking the Impact of Hurricane Helene: Power Outages and Public Sentiment

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Introduction

Hurricanes are among the most destructive natural disasters, bringing not only immediate devastation but also long-lasting social and economic consequences. Hurricane Helene severely impacted communities across the southeastern United States, causing widespread power outages and infrastructure damage, along with affecting public emotions

However, the true impact of such storms extends beyond physical damage. Prolonged blackouts disrupt healthcare, businesses, and daily life—while also triggering emotional responses like fear, anger, and frustration.

By analyzing how outages, and public sentiment evolved throughout the storm, we aim to better understand how communities experience and respond to hurricanes—offering insights that can improve disaster preparedness, emergency communication, and post-disaster recovery.

Research Goal

To analyze how Hurricane Helene affected communities by examining:

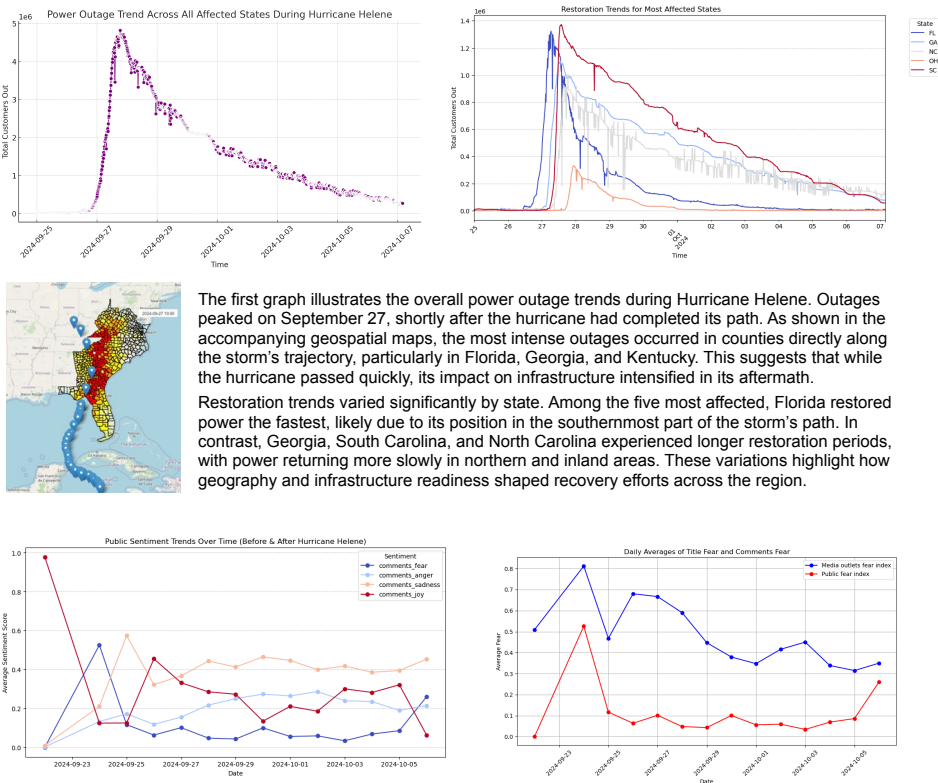
- The scale and duration of power outages
- Shifts in public sentiment before, during, and after the storm
- The relationship between outages and public sentiment

This study combines geospatial outage data, sentiment analysis, and data visualization to uncover patterns in public response and storm impact.

Methods

- **Power Outages:**
We collected county-level power outage data from September 26 to October 8 across impacted states. This allowed us to map outage intensity, track restoration timelines, and identify the most affected counties and states during the hurricane.
- **Public Sentiment Analysis:**
Using Natural Language Processing, we analyzed YouTube video titles and comments to assess public emotion. Sentiment categories included fear, sadness, anger, and joy. This analysis revealed how public concern fluctuated in response to Helene’s progression and infrastructure disruptions.

Figures and Results



The first graph illustrates the overall power outage trends during Hurricane Helene. Outages peaked on September 27, shortly after the hurricane had completed its path. As shown in the accompanying geospatial maps, the most intense outages occurred in counties directly along the storm’s trajectory, particularly in Florida, Georgia, and Kentucky. This suggests that while the hurricane passed quickly, its impact on infrastructure intensified in its aftermath.

Restoration trends varied significantly by state. Among the five most affected, Florida restored power the fastest, likely due to its position in the southernmost part of the storm’s path. In contrast, Georgia, South Carolina, and North Carolina experienced longer restoration periods, with power returning more slowly in northern and inland areas. These variations highlight how geography and infrastructure readiness shaped recovery efforts across the region.

This graph illustrates public sentiment trends over time during Hurricane Helene, based on YouTube titles and comments. Fear peaked two days before landfall, with both titles and comments reflecting elevated anxiety. Notably, titles showed higher levels of fear than comments, suggesting that media framing may have amplified public concern in the days leading up to the storm.

After landfall, fear sentiment dropped sharply and remained low, while sadness rose and stayed consistently high, aligning with prolonged power outages and storm-related hardship. Simultaneously, anger steadily increased, likely reflecting mounting frustration with delayed recovery efforts and perceived failures in infrastructure and emergency response.

These trends reveal a clear emotional shift throughout the event—from anticipatory fear before landfall to persistent sadness and growing frustration during the recovery period.

Conclusion

This study explored the multifaceted impact of Hurricane Helene by analyzing power outages and public sentiment across affected communities. Our findings reveal that infrastructure failure and emotional response were deeply intertwined during and after the storm.

Power outages peaked on September 27, just after the hurricane’s path had moved through the region, affecting millions of residents across states like South Carolina, Georgia, and Florida. Restoration timelines varied widely, with Florida recovering the fastest and more northern states experiencing extended blackouts—highlighting geographical and infrastructural differences in emergency response.

Public sentiment, captured through YouTube comments and titles, mirrored these disruptions. Fear surged before landfall, largely driven by media narratives, while sadness and anger grew steadily post-landfall, reflecting the emotional toll of widespread outages and slow recovery efforts.

Together, these insights demonstrate how communities experience and react to disasters not just physically, but emotionally. By integrating geospatial outage data with real-time sentiment analysis, this research provides a powerful framework for understanding public distress and evaluating the effectiveness of disaster response. These tools can help inform future emergency planning, guide more targeted communication strategies, and promote resilience in the face of increasingly frequent climate-related events.

Acknowledgments

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References

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